

Quick Check

Lesson 1: The Farmer's Puzzle: Why Do Maize and Beans Behave Differently After the Same Rain?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 1

Mama Wanjiru in Molo plants maize and beans on the same morning after the first heavy rains. The seeds all look healthy. After one week, which description matches what the story says she found?

- A** All the seeds came up on the same day
- B** Every healthy-looking seed had sprouted
- C** Only the beans came up and every maize seed failed
- D** Some seeds had sprouted, some were late, and a few had not come up at all

While watching a time-lapse of a bean seed growing in soil, four learners wrote notes. Which note is a specific observation, something you can actually see, and not a guess or an explanation?

- A** The bean seed needs warmth to wake up.
- B** Maize and beans grow in exactly the same way.
- C** The bean stem was bent like a hook as it came above the soil.
- D** The plant grew.

In the time-lapse videos, learners were asked to watch which part of the seedling came out first. Which statement is an 'I wonder' statement rather than an 'I notice' statement?

- A** I notice that the maize leaf was rolled up.
- B** I wonder why some seeds that looked healthy did not sprout.
- C** I notice that some seeds did not sprout.
- D** I notice that the root grew down into the soil.

Your teacher says your first drawn model of a maize seed and a bean seed will not be marked right or wrong, and you will keep it in a folder. What is the main reason for this?

- A** It shows your starting ideas, so you can compare them with your improved ideas at the end of the unit
- B** It will be used to give each learner a final grade for the term
- C** Only learners who draw the correct model are allowed to continue
- D** The teacher already knows that every first model is wrong and will discard them

On Day 1 of the unit, a learner asks: 'Why did some of Mama Wanjiru's healthy-looking seeds not germinate?' Which is the best response at this stage of learning?

- A** Only the biggest seeds germinate, so the small ones failed.
- B** Healthy-looking seeds always germinate, so the farmer must have counted wrongly.
- C** It has not been explained yet, so it goes on the Driving Question Board to investigate with evidence.
- D** Seeds need only water, so the rain should have made every seed germinate.

A group wants to post a question on the Driving Question Board about maize and beans. Which question is the best one to investigate?

A How does maize grow differently from beans?

B Is maize good?

C Is Mama Wanjiru a good farmer?

D Did it rain in Molo?

Many groups asked whether growth and development are the same thing. Which question did the class agree to investigate first in the next lesson?

A Why is the rainfall heavy in Nakuru County?

B What is growth and development?

C Which fertiliser is best for maize?

D How do plants make their own food?

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).